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Sensitivity Analysis of Subsurface Hydrogen and Thermal Energy Storage with Insights from a Review of Techno-Economic and Life-Cycle Assessments

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Abstract

Renewable energy is gaining momentum as the answer for the world greenhouse gas (GHG) emission issues. However, the unpredictable nature of wind and solar energy puts immense pressure on the electric grid, leading to substantial energy losses, as seen in Texas. While storage technologies hold promises in addressing this challenge, their integration remains limited due to inefficiencies and an incomplete understanding of their impact, highlighting the need for further study and development.

This paper reviews the results and approaches of existing techno-economic analysis (TEA) and life-cycle assessments (LCA) of two subsurface energy storage technologies: subsurface hydrogen (H₂) storage and geothermal energy storage, while addressing the gap in TEA and LCA. In addition, this study integrates TEA and LCA frameworks to perform a sensitivity analysis, evaluating the influence of various factors across multiple phases—including production, processing withdrawal, and power generation—on simulation outcomes, overall efficiency, cost, and environmental impacts.

The review revealed that while TEA remains the dominant approach, LCA is increasingly recognized as a critical complement—offering a more comprehensive understanding of environmental impacts throughout a system's life span. Most hydrogen-related studies continue to focus on the production phase, particularly via electrolysis, with limited attention to subsurface storage in porous formations such as aquifers. Similarly, geothermal research is primarily centered on electricity generation, while aquifer thermal energy storage (ATES) is often studied only for heating and cooling applications, not power production. As a result, there is a noticeable gap in integrated TEA–LCA assessments specifically targeting long-duration subsurface energy storage technologies in porous media. In addition, the conducted probabilistic and one-at-a-time (OAT) sensitivity analyses revealed how variations in curtailed energy availability, electrolyzer efficiency, and reservoir performance parameters influence system efficiency, levelized cost, and greenhouse gas emissions.

These findings emphasize the need for more site-specific modeling and suggest that subsurface hydrogen storage may offer improved efficiency when key parameters are optimized. The sensitivity analysis adds further insight by quantifying how critical variables influence system performance. Together with the

literature review, this study contributes to a more informed basis for evaluating long-duration subsurface storage technologies.

Introduction

Electricity generation from renewable sources such as wind and solar is essential for reducing greenhouse gas emissions and lowering dependence on fossil fuels. However, the variability of sunlight and wind presents challenges for maintaining a stable and reliable power supply. According to the International Renewable Energy Agency (IRENA, 2020a), the global shift toward renewable energy has gained significant momentum in recent years, with projections suggesting that renewables could supply up to 86% of the world's electricity by 2050. In the United States, solar and wind capacity has expanded rapidly over the past decade. However, this expansion has also resulted in a noticeable rise in energy curtailment—particularly in areas with high renewable generation, such as the Electric Reliability Council of Texas (ERCOT). To mitigate this challenge, energy storage technologies are considered a promising solution.

Energy storage technologies are commonly categorized into four main types: electrochemical, thermal, mechanical, and electrical. Each category has unique properties—such as discharge time and power output—that influence its suitability for specific applications within the power grid (IRENA, 2017). According to Argyrou et al. (2016), the selection of an appropriate storage system depends on multiple factors, including cost, environmental impact, storage capacity (ranging from a few kilowatt-hours to several gigawatt-hours), and the desired duration of energy storage (ranging from minutes to several weeks or months).

Subsurface energy storage has gained considerable attention in recent years due to its capacity to store larger volumes of energy compared to traditional methods (Chen et al., 2023), along with its relatively low environmental and social impact. Two main types of geological formations are utilized for large-scale subsurface storage: engineered cavities and porous media. Engineered cavities are man-made underground chambers, typically created by excavating hard or soft rock formations or by dissolving salt deposits to form salt caverns. On the other hand, porous media takes advantage of naturally occurring voids within sedimentary rock formations, such as aquifers and depleted hydrocarbon reservoirs, which offer the porosity and permeability needed for effective fluid storage.

Several long-duration energy storage solutions are being explored in response to growing concerns about the stability of renewable energy supply, including subsurface H₂ storage and synthetic geothermal storage. Subsurface H₂ storage involves converting surplus renewable electricity into green hydrogen through electrolysis, a zero-emission process that splits water into hydrogen and oxygen. The resulting hydrogen is then injected into subsurface formations such as saline aquifers or depleted oil and gas reservoirs. According to Chen et al. (2023), geologic structures like domes or anticlines, sealed by impermeable caprock, are particularly favorable for hydrogen storage as they help retain the gas and minimize the production of formation water during retrieval. This stored hydrogen can later be used for a variety of applications, including power generation, industrial use, transportation, heating, or as a feedstock for synthetic fuels like ammonia.

On the other hand, synthetic geothermal storage aims to store thermal energy derived from solar sources. Solar energy may be captured in the form of electricity through photovoltaic (PV) panels or as heat via solar thermal systems. PV-generated or wind generated electricity can be used to operate electric boilers that produce steam, while solar thermal collectors heat fluids directly. This thermal energy is then stored underground and later extracted to generate electricity during periods of low energy resources, thus enhancing system reliability.

To fully assess the viability of these emerging subsurface energy storage technologies, it is essential to evaluate their technical performance, along with associated economic costs and environmental impacts. This paper reviews existing TEA and LCA related to subsurface H₂ storage and synthetic geothermal storage, emphasizing key methodologies and findings. Additionally, it presents a sensitivity analysis to quantify how

variations in different parameters—such as energy input, processing requirements, and subsurface injection and withdrawal conditions—affect overall system efficiency, levelized cost, and greenhouse gas emissions.

Methodology

Monthly wind curtailment data (in MWh) from the ERCOT region, reported for the years 2017 to 2021 (Hoza, 2022), was used to conduct a sensitivity analysis, evaluating the impact of key variables on the total system efficiency of both subsurface H₂ and synthetic geothermal energy storage systems. Following a methodology similar to that of Tayyib et al. (2025), the output and intermediate equations—outlined in Table 1—capture the major stages of each system considered in the analysis. For the hydrogen system, the stages include: Production (H₂ generation via electrolysis), Processing (electricity consumption for compression and pumping), Storage (injection into subsurface aquifers, accounting for withdrawal volumes), and Power Generation (conversion of stored hydrogen back into electricity via fuel cells). For the synthetic geothermal system, the stages are: Thermal Energy Production (using electric-powered boilers), Processing (water heating, pressurization, and pumping), Storage (injection into subsurface reservoirs), and Power Generation (electricity generation using turbine-generator systems).

Table 1—Summary of equations used for the sensitivity analysis by stage, adapted from Tayyib et al. (2025).

Output Equation	
System's Total Efficiency [%]	$\eta_{\text{system}} = \frac{\text{Power Generated}}{(\text{Curtailed Energy} + \text{Electrical Energy Consumption})} \times 100\%$
Intermediate Equations	
Production Stage	For H₂: $m_{\text{H}_2 \text{ Produced}} [\text{Kg}] = \text{Curtailed Energy} \div \eta_{\text{electrolyser}}$
	For Geothermal: $Q = C_v m_{\text{water}} \Delta T$ $Q_{\text{total}} = Q_1 + Q_2 + Q_3$ $m_{\text{water}} [\text{Kg}] = \frac{Q_{\text{total}}}{C_{p1} \Delta T1 + H + C_{p2} \Delta T2}$
Processing Stage	For H₂: $\text{Energy Required for Compression} \left(\frac{E}{Q_{\text{in}} \times \tau \times \rho_{\text{H}_2}} \right) [\text{kWh}] = \left(\frac{2 \times 10^{-5}}{\gamma \times \rho_{\text{H}_2}} \times \left[\left(\frac{P_{\text{in}}}{P_{\text{out}}} \right)^{\gamma} - 1 \right] \right) \times \frac{277 \text{ kWh}}{\text{GJ}}$ $\text{Energy Required for Pumping (P)} [\text{kWh}] = \eta_p \rho_{\text{H}_2} V_{\text{H}_2} g h \times \frac{1 \text{ N}}{\text{kg} \cdot \frac{\text{m}}{\text{s}^2}} \times \frac{W}{\frac{\text{mN}}{\text{s}}} \times \frac{1 \text{ MW}}{10^6 \text{ W}} \times 24 \text{ hr}$
	For Geothermal: $\text{Energy Required for Pumping (P)} [\text{MWh}] = \eta_p \rho_{\text{water}} V_{\text{H}_2 O} g h \times \frac{1 \text{ N}}{\text{kg} \cdot \frac{\text{m}}{\text{s}^2}} \times \frac{W}{\frac{\text{mN}}{\text{s}}} \times \frac{1 \text{ MW}}{10^6 \text{ W}} \times 24 \text{ hr}$ $\text{Energy Required for Heating (for Electric Boiler)} [\text{MWh}] = m_{\text{water}} \times \text{Req}_{\text{Heat}}$ $\text{Energy Required for Condensation (for CSP)} [\text{MWh}] = m_{\text{water}} \times Q_{\text{cond}}$
Storage Stage	For H₂: $\text{Injection Volume} \left[\frac{\text{m}^3}{\text{d}} \right] = m_{\text{H}_2 \text{ produced}} \div \rho_{\text{H}_2} \text{ at specific pressure \& temperature}$ $\text{Temperature at a specific depth } [^{\circ}\text{C}] = T_{\text{surf}} + \frac{\Delta T}{\Delta D} \times D$ $\text{Withdrawal Volume} \left[\frac{\text{m}^3}{\text{d}} \right] = \text{Injection Volume} \times \text{Withdrawal Efficiency}$ $\text{Withdrawal Mass} [\text{Kg}] = \text{Withdrawal Volume} \times \rho_{\text{H}_2 \text{sub}}$
	For Geothermal: $\text{Injection Volume (V}_{\text{H}_2\text{O}}) \left[\frac{\text{m}^3}{\text{d}} \right] = \frac{m_{\text{water}}}{\text{month}} \times \frac{1 \text{ month}}{\text{number of Days}} \div \rho_{\text{water}}$ Based on Reservoir simulation: $\text{Withdrawal Efficiency} [\%] = \frac{Q_{\text{out}}}{Q_{\text{in}}} = \frac{m_{\text{out}} \times (T_{\text{out}} - T_{\text{surf}})}{m_{\text{in}} \times (T_{\text{in}} - T_{\text{surf}})}$ $\text{Withdrawal Volume} \left[\frac{\text{m}^3}{\text{d}} \right] = \text{Injection Volume} \times \text{Withdrawal Efficiency}$
Energy Production Stage	For H₂: $\text{Energy Produced} [\text{MWh}] = \text{Withdrawal mass} \times \text{LHV}$
	For Geothermal: $\text{Withdrawal Mass} [\text{Kg}] = \text{Withdrawal Volume} \left[\frac{\text{m}^3}{\text{d}} \right] \times \frac{\text{No. of Days}}{1 \text{ Month}} \times \rho_{\text{water}}$ $\text{Power Produced} [\text{MWh}] = \text{Withdrawal Mass} [\text{Kg}] \times H \times \eta_{\text{turb}} \times \eta_{\text{gen}}$

To evaluate how variations in input data affect total system efficiency, a Monte Carlo-based sensitivity analysis was conducted for both storage systems. This probabilistic method accounts for uncertainty in input values—arising from natural variability, measurement error, or unknown future conditions—by repeatedly sampling from predefined probability distributions. The input variables were modeled using triangular distributions, defined by minimum, maximum, and most likely (average) values. These ranges, along with the rationale for including each variable in the analysis, are summarized in Table 2. The simulation was run for 10,000 iterations, enabling a robust statistical assessment of how each parameter influences system efficiency. The results were then used to quantify and rank the contribution of each variable to output variance, highlighting the most influential factors in each system.

Table 2—Summary of independent input variables, distributions, and sources used for the analysis.

Parameter	Reason for Inclusion	Unit	Range (Min–Max–Avg)	Data Reference
Curtailed Energy	Varies with grid demand and renewable generation; impacts losses and efficiency.	MWh	62k – 538k – 314k	Hoza, 2022 (ERCOT Data, 2017–2021)
Electrolyzer Efficiency (η_e)	Depends on technology, age, and operating conditions; affects hydrogen output.	kg H ₂ /MWh	12 – 20 – 16	IRENA, 2020b
Injection Pressure	Affects compression energy and storage; varies with geology and operations.	bar	50 – 300 – 175	Barthelemy et al., 2021
Injection Depth	Site-specific; affects pumping energy and temperature profile.	m	500 – 3000 – 1750	Shi et al., 2023; Yu et al., 2013
Pump Efficiency (η_p)	Affects pumping energy requirements; depends on equipment and fluid type.	%	50 – 80 – 65	Kilgore, 2024 (H ₂); Shaw, 2022 (water)
Withdrawal Efficiency	Influences recoverable energy; dependent on design and reservoir properties.	%	H ₂ : 75 – 90 – 83 Geo: 33 – 75 – 54	Okoroafor et al., 2022 (H ₂); Tayyib et al., 2024 (Geo)
Turbine Efficiency	Impacts power conversion; varies with turbine design and condition.	%	85 – 90 – 88	Wagner, 2011; Beckers & Johnston, 2022
Generator Efficiency	Affects electrical output and total system efficiency.	%	80 – 90 – 85	Islam, 2022

While Monte Carlo analysis offers a comprehensive probabilistic evaluation of how uncertainties across multiple input variables collectively affect system performance, a complementary local sensitivity analysis was also performed. This one-at-a-time (OAT) approach involved varying each key parameter individually using the maximum, minimum, and average values reported in the literature, while holding all other inputs constant at their average values. This technique isolates the direct effect of each parameter on system efficiency, providing clearer insights into which variables exert the most significant individual influence.

Following the sensitivity analyses, a comprehensive review of the literature was conducted, focusing on existing TEA and LCA for both technologies. This review aimed to compare methodologies and outcomes, extract key insights, and identify knowledge gaps, enhancing the understanding of the current state of TEA and LCA for subsurface hydrogen and synthetic geothermal energy storage.

Results and Discussions

Curtailment Data

Fig. 1a displays the monthly curtailment data in ERCOT for both wind and solar energy, with peaks notably occurring in the spring months—particularly in March and April of 2018 and 2021. However, since solar curtailment is relatively low compared to wind, the analysis focuses primarily on wind curtailment, as it presents a more critical case for the integration of energy storage solutions. The trend is further supported by the aggregated monthly averages in Fig. 1b, which show consistently high wind curtailment during March, April, October, and November, and minimal levels during the summer months—especially June through August.

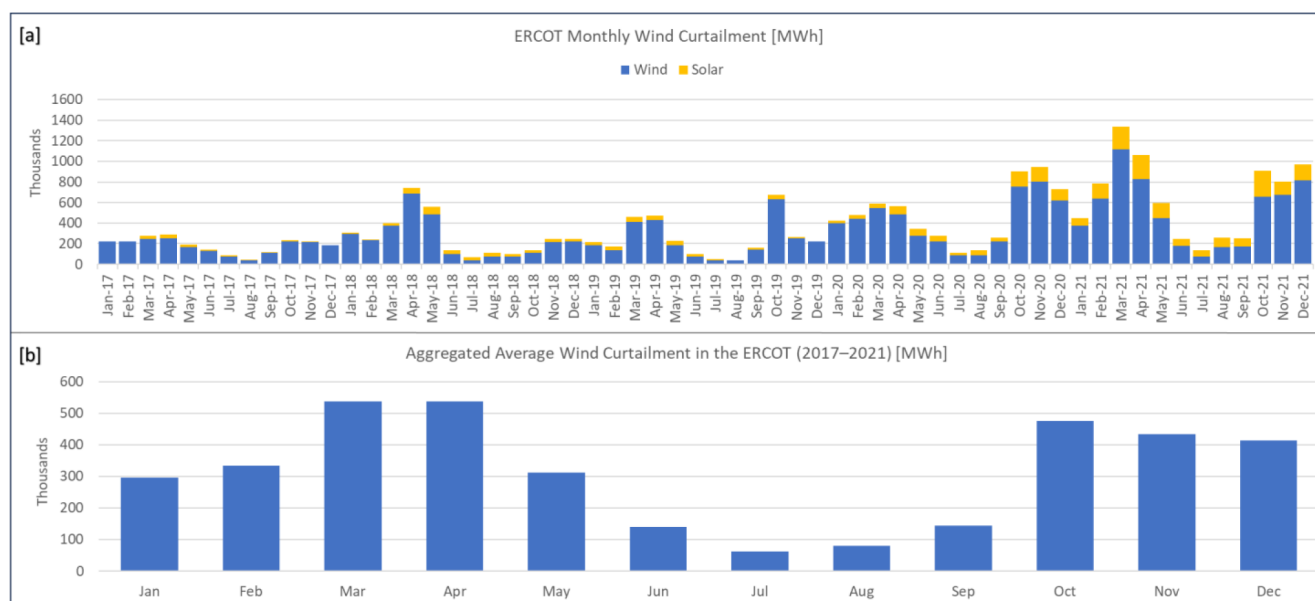


Figure 1—ERCOT curtailment: (a) monthly wind and solar curtailment from 2017 to 2021; (b) aggregated average monthly wind curtailment across the five-year period. Graphs generated using data from [Hoza \(2022\)](#).

These observed patterns are sensitive to seasonal wind speeds and grid demand. Fig. 2 shows that wind speeds across Texas are highest in the spring, often exceeding 10 mph, which corresponds to the same months with the highest curtailment. Meanwhile, Fig. 3 demonstrates that electricity demand in the U.S. is lowest in spring and fall, when temperatures are milder and heating/cooling needs are reduced, and highest in the summer (July–August), when air conditioning drives up electrical consumption. This creates conditions where wind generation exceeds what the grid can absorb during low-demand months, leading to increased curtailment. In contrast, during summer months, higher demand and lower wind speeds result in better alignment between generation and consumption, thereby minimizing curtailment. This suggests that curtailment in ERCOT is influenced by both wind availability and seasonal imbalances between electricity supply and demand, highlighting the need for flexible, seasonal energy storage solutions—such as subsurface H₂ or synthetic geothermal systems—to store excess energy during high-curtailment periods and discharge it during high-demand seasons, improving overall system efficiency and renewable utilization.

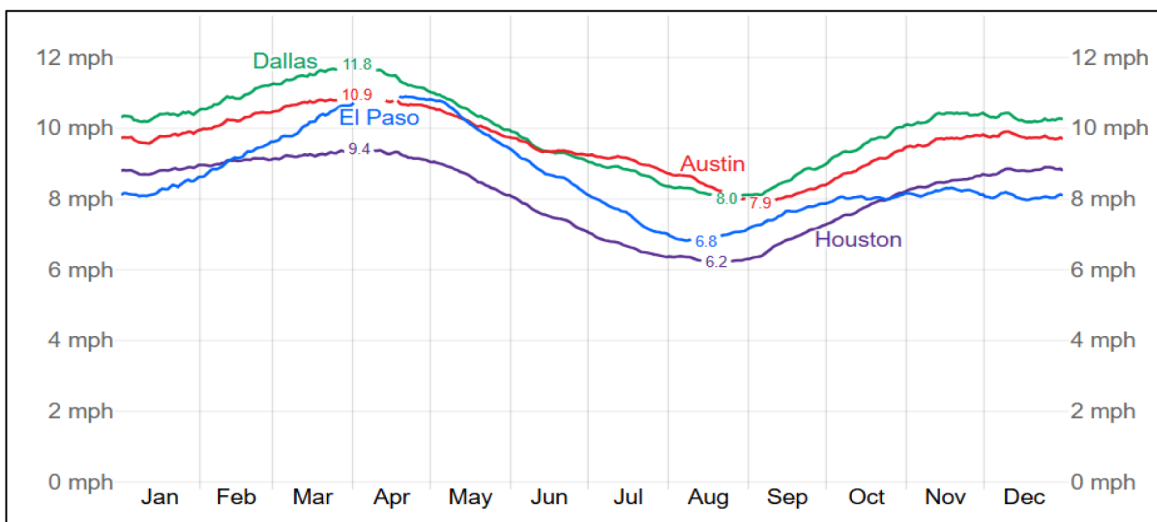


Figure 2—Monthly average wind speed in Texas (Weather Spark, 2016).

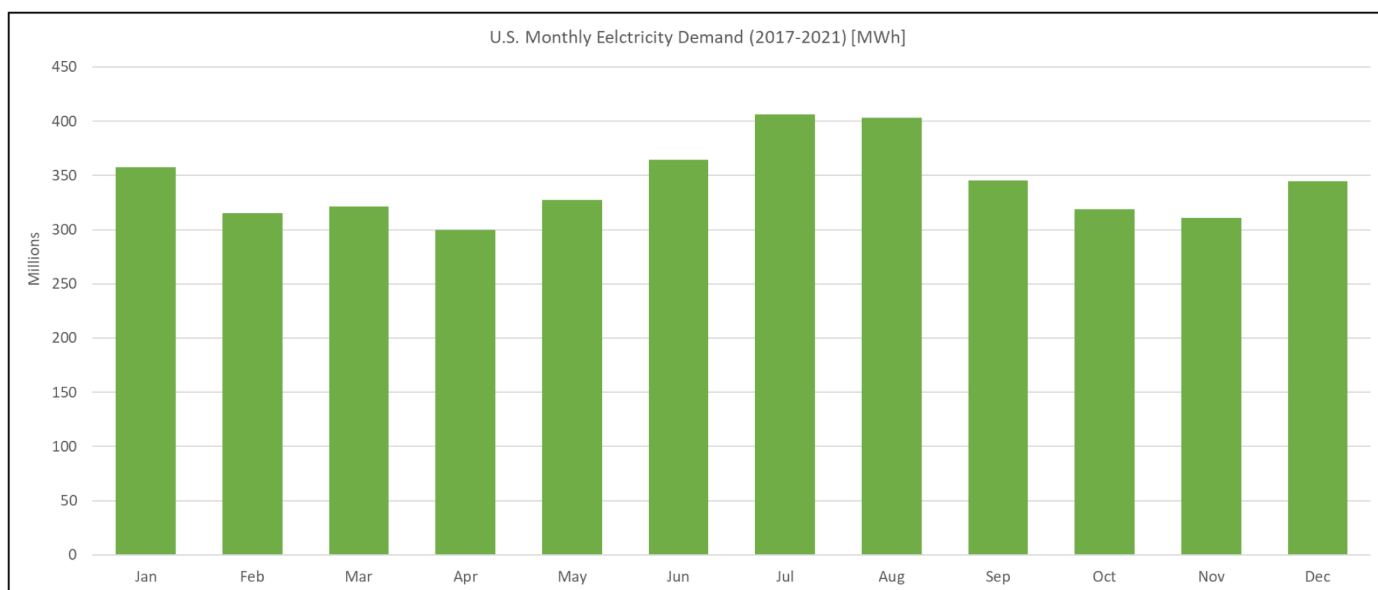


Figure 3—Monthly electricity demand in Texas (Ember, 2025).

Sensitivity Analysis

Monte-Carlo Simulation. Fig. 4 presents a sensitivity assessment of the total efficiency of the subsurface H₂ storage system, based on a Monte Carlo simulation with 10,000 iterations. As shown in Fig. 4a, the distribution of system efficiency reveals that most outcomes fall within the range of around 0.40 to 0.50, indicating a consistent performance under varying input conditions. The 5th and 95th percentiles—at 0.37 and 0.53 respectively—represent a probabilistic boundary for expected operational performance, confirming that the system is capable of achieving moderate to high efficiency under a variety of input conditions. Fig. 4b ranks the parameters with the greatest influence on system efficiency. Curtailment was identified as the most influential factor affecting the total efficiency of the subsurface H₂ storage system, which aligns with the principle that the volume of surplus renewable energy available directly limits the amount of hydrogen that can be produced and stored. While this may seem intuitive, the sensitivity analysis quantifies its dominant influence compared to other parameters. Following curtailment, injection volume and withdrawal efficiency were the next most impactful parameters. This indicates that the effectiveness

of the storage reservoir—both in its capacity to receive injected hydrogen and in minimizing losses during withdrawal—is just as important as the electrochemical processes involved in hydrogen production. Electrolyzer efficiency, injection pressure, and injection depth showed smaller, but still notable, effects. Ultimately, this sensitivity analysis provides valuable insight into which technical levers are most impactful for maximizing the effectiveness of subsurface hydrogen storage.

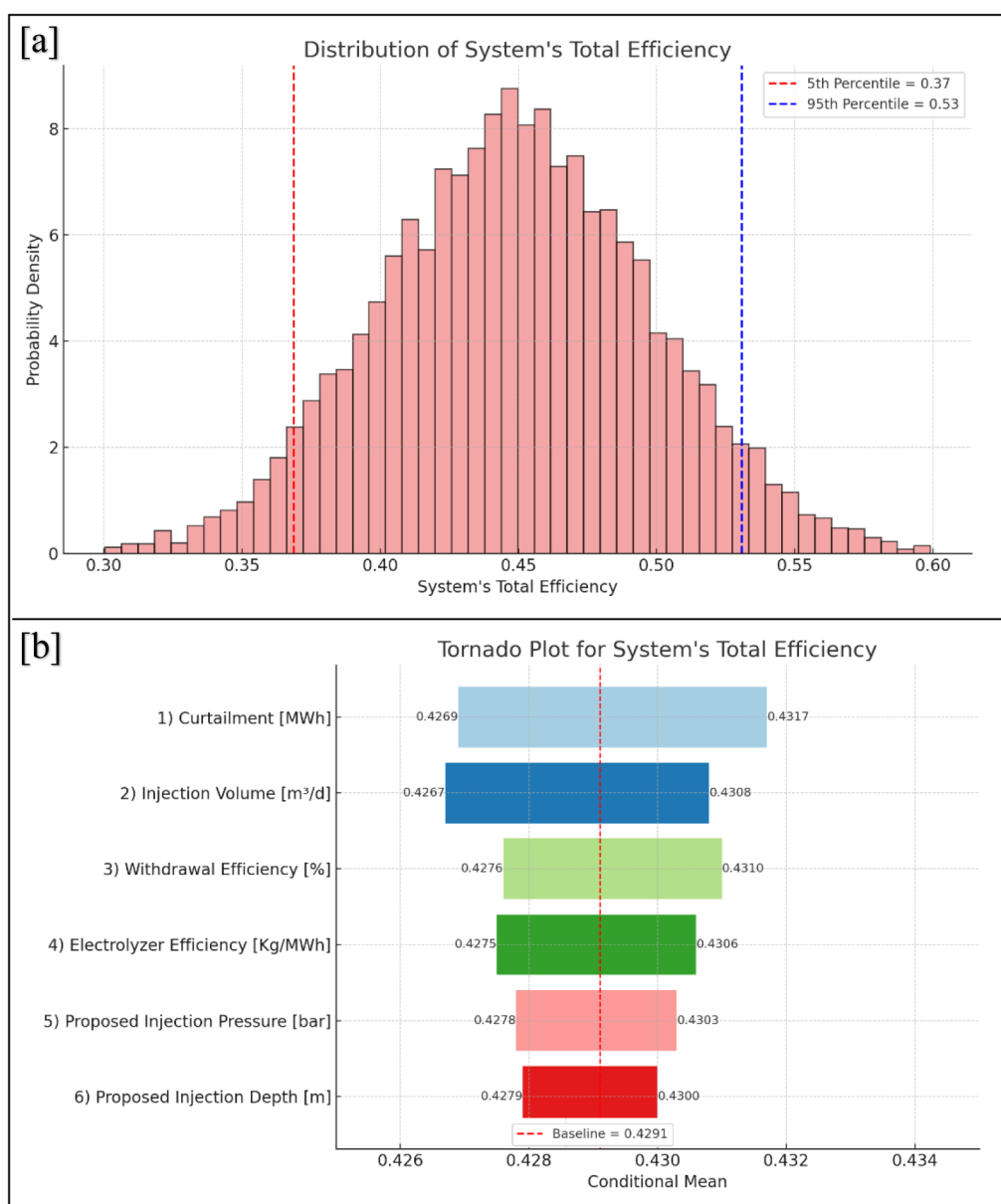


Figure 4—Monte Carlo simulation results for subsurface H₂ storage system: (a) distribution of total system efficiency and (b) sensitivity analysis using tornado plot.

Fig. 5 presents a sensitivity assessment of the total efficiency of a synthetic geothermal energy storage system powered by an electric boiler. In Fig. 5a, the Monte Carlo simulation results show that the system's total efficiency predominantly falls between 0.120 and 0.153, with a mean around 0.1334. The relatively narrow distribution suggests consistent but inherently low performance, reflecting the limitations of electric boilers in large-scale thermal storage. Fig. 5b further ranks the influence of input parameters using a tornado plot. Generator and turbine efficiencies clearly dominate, indicating that optimizing electricity-to-heat and heat-to-electricity conversion processes would yield the greatest improvements in overall system

performance. In contrast, parameters such as pump efficiency, injection pressure, injection depth, and withdrawal efficiency had negligible impact, implying that subsurface and fluid-handling operations are not major constraints in this configuration. These findings suggest that future investment and innovation should prioritize enhancing conversion technologies rather than modifying subsurface or mechanical components.

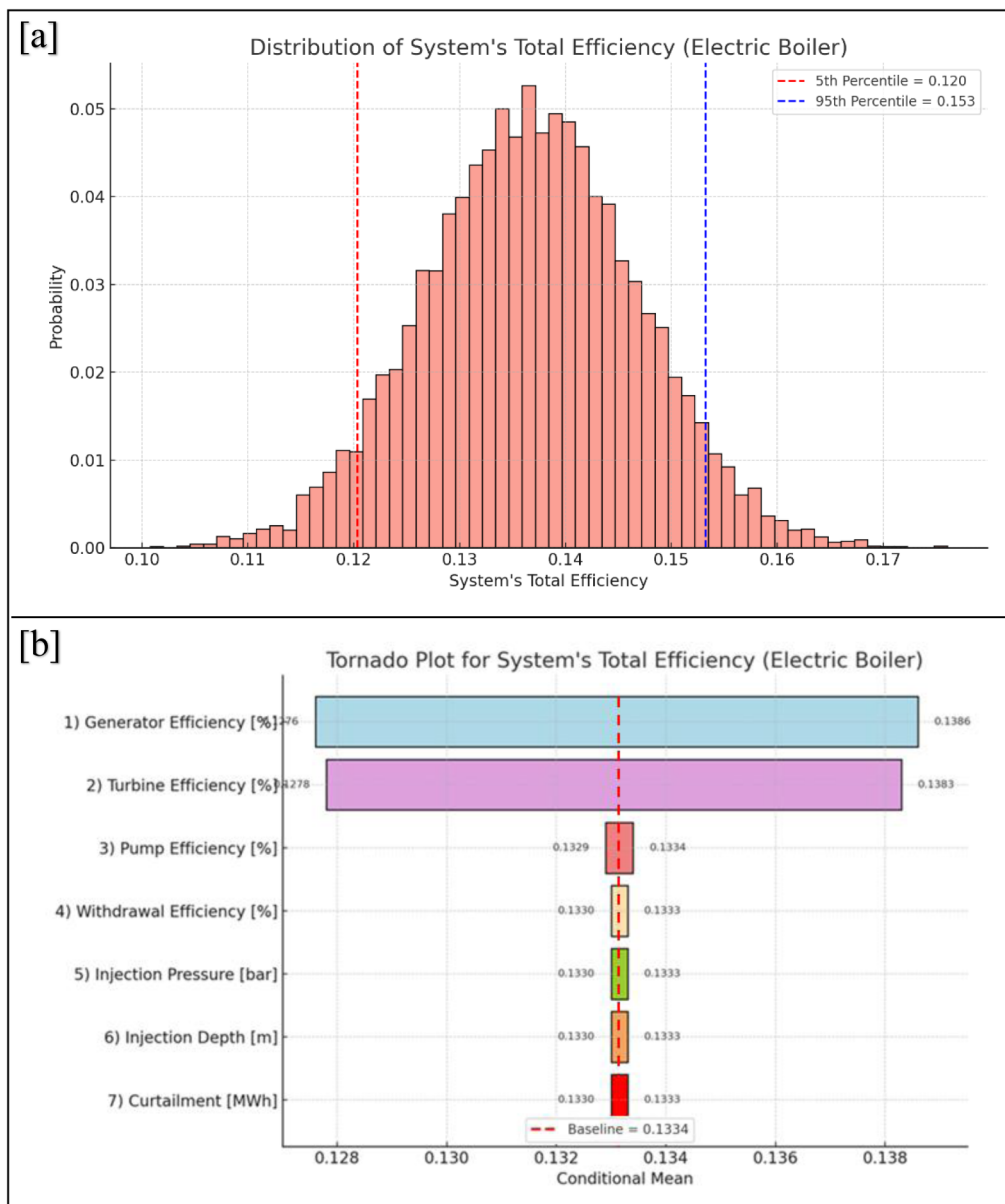


Figure 5—Monte Carlo simulation results for synthetic geothermal storage system: (a) distribution of total system efficiency and (b) sensitivity analysis using tornado plot.

One-at-a-Time (OAT) Sensitivity Analysis. Table 3 presents the results of the OAT analysis on subsurface H₂ storage parameters, conducted by individually varying each input within a realistic range based on operational and equipment constraints reported in the literature. The results show that electrolyzer efficiency has the most significant influence, with a $\pm 25\%$ variation leading to a $\pm 24\%$ change in system efficiency. Withdrawal efficiency also exhibits a measurable impact, shifting the efficiency by approximately $\pm 8\%$. In contrast, changes in injection pressure, injection depth, and pump efficiency showed minimal influence on total system efficiency, with shifts limited to $\pm 1\%$, indicating that mechanical parameters contribute little to the overall energy performance.

Table 3—OAT sensitivity analysis for subsurface H2 storage system (Reported as % change from baseline)

Parameters	Input Value	H2 Production [KgH2]	Compression Requirement [MWh]	Pumping Requirement [MWh]	Generated Energy [MWh]	System's Efficiency [%]	Levelized Cost [\$ /MWh]	Emission [Tons]
Base		5024000	4099	3683	138858	43.15%	124	37513
Curtailed Energy	Min	-80%	-80%	-80%	-80%	0%	64.44%	-80.25%
	Max	71%	71%	71%	71%	0%	-6.60%	71.34%
Electrolyzer Efficiency	Min	-25%	-25%	-25%	-25%	-25%	5.28%	-25.00%
	Max	25%	25%	25%	25%	24%	-3.17%	25.00%
Injection Pressure	Min	0%	-39%	0%	0%	1%	0%	0%
	Max	0%	22%	0%	0%	0%	0%	0%
Injection Depth	Min	0%	0%	-71%	0%	1%	0%	0%
	Max	0%	0%	71%	0%	-1%	0%	0%
Pump Efficiency	Min	0%	0%	-23%	0%	0%	0%	0%
	Max	0%	0%	23%	0%	0%	0%	0%
Withdrawal Efficiency	Min	0%	0%	0%	-10%	-10%	1.69%	-9.64%
	Max	0%	0%	0%	8%	8%	-1.23%	8.43%

The levelized cost shows the highest sensitivity to the amount of available curtailed energy. Reducing curtailed energy by 80% leads to a 64.44% increase, while increasing curtailed energy by 71% reduces levelized by 6.60%. Higher availability of excess renewable energy allows the storage system to operate at higher capacity, improving economic performance by spreading fixed and capital costs across more produced energy. Electrolyzer efficiency is also a key driver of cost. A 25% decrease in efficiency increases the levelized cost by 5.28%, while a 25% increase reduces it by 3.17%. Other parameters, such as withdrawal efficiency, show a smaller but noticeable impact on cost.

Following the analysis of subsurface hydrogen storage, Table 4 presents a similar OAT sensitivity assessment for the synthetic geothermal storage system. Each parameter was varied within realistic bounds based on equipment capabilities and operating conditions reported in the literature. Reducing curtailed energy by 80% increased levelized cost by 24.9%, while a 71% increase decreased it by only 2.5%. GHG emissions followed a similar pattern, varying from -78.8% to +70.0%, due to the correlation with the volume of energy processed. Withdrawal efficiency had a significant effect on performance, with a $\pm 39\%$ change in generated energy and system efficiency, reflecting the importance of effective heat recovery from the storage formation. Turbine and generator efficiencies also impacted system performance, particularly emissions. Turbine efficiency changes led to $\pm 3\%$ shifts in system efficiency and $\pm 38.2\%$ variation in emissions, while generator efficiency resulted in $\pm 6\%$ changes in system efficiency and $\pm 3.3\%$ shifts in emissions. These results highlight the downstream impact of energy conversion efficiency on overall system performance and life-cycle emissions.

Table 4—OAT sensitivity analysis for subsurface Hydrogen storage system (Reported as % change from baseline)

Parameters	Input Value	Heated Water Mass [Kg]	Heating Requirement [MWh]	Pressurizing Requirement [MWh]	Pumping Requirement [MWh]	Generated Energy [MWh]	System's Efficiency [%]	Levelized Cost [\$ /MWh]	Emission [Tons]
Base		897627291	628339	0	8	77860	8.26%	489	1.031
Curtailed Energy	Min	-80%	-80%	-80%	-80%	-80%	0%	24.9%	-78.8%
	Max	71%	71%	71%	71%	71%	0%	-2.5%	70.0%
Injection Depth	Min	0%	0%	0%	-71%	0%	0%	0%	0%
	Max	0%	0%	0%	71%	0%	0%	0%	0%
Pump Efficiency	Min	0%	0%	-23%	-23%	0%	0%	0%	0%
	Max	0%	0%	23%	23%	0%	0%	0%	0%
Withdrawal Efficiency	Min	0%	0%	0%	0%	-39%	-39%	0%	0%
	Max	0%	0%	0%	0%	39%	39%	0%	0%
Turbine Efficiency	Min	0%	0%	0%	0%	-3%	-3%	0%	-38.2%
	Max	0%	0%	0%	0%	2%	2%	3.9%	38.2%
Generator Efficiency	Min	0%	0%	0%	0%	-6%	-6%	0.2%	-3.3%
	Max	0%	0%	0%	0%	6%	6%	-0.1%	2.2%

TEA and LCA Review

Table 5 provides an overview of recent studies applying TEA and LCA across a variety of hydrogen production, storage, and geothermal energy systems. While TEA remains the dominant methodology—reflecting continued interest in the economic feasibility of emerging energy technologies—an increasing number of studies incorporate LCA to evaluate environmental impacts. Notably, several studies enhance these core assessments with complementary methodologies such as sensitivity analysis (e.g., Ganguly (2025), Sadeghi (2020), Yousefi (2023)), Monte Carlo simulations (Yousefi (2023), Schuppler (2019), Stemmler (2021)), and optimization modeling (Aviles (2021), Malek (2022)), ensuring that both economic and environmental evaluations consider system variability and uncertainty. Studies like those by Bareiß (2021), Brandt (2023), and Ganguly (2025) offer insights into emission factors.

Table 5—Summary of Recent TEA and LCA Studies on Hydrogen and Geothermal Systems

Reference	System Type	Methodology	Findings	Location
Titus et al. (2025)	Geothermal + BECCS/DACCS	TEA, LCA	- BECCS offers lowest cost of abatement (\$122/tCO₂) down to \$45/tCO₂ retrofitted) - DACCS more costly (\$187/tCO₂) - BECCS provides more power (511 MWe vs 190 MWe for DACCS)	Italy
Ganguly et al. (2025)	Biohydrogen from waste	TEA, LCA, Sensitivity Analysis	- Emissions of -8.6 and -8.0 kg CO₂-eq/kg H₂ (CW and SFW) - Cost ranges \$4-7/kg H₂ at high current density	Europe
Yousefi et al. (2023)	Underground hydrogen storage using hydrogen and nitrogen cushion gas	TEA, LCA, Monte Carlo Analysis,	- Hydrogen cushion gas cost and interest rates are key cost drivers - Nitrogen reduces LCOHS - Economic viability improves with optimized design and financial incentives	Australia
Shahabuddin et al. (2023)	Alkaline electrolysis hydrogen production using solar, wind, and hybrid systems	TEA, LCA, Sensitivity Analysis	- Electricity cost is the largest contributor to LCH₂ - Future reductions expected due to higher capacity factors and lower electrolyzer CAPEX - Region-specific results emphasize need for tailored approaches.	Global (modeled)
Brandt (2023)	Geologic Hydrogen	LCA, Sensitivity Analysis, Modeling (OPGEE)	- GHG intensity as low as 0.4 kg CO₂-eq/kg H₂ - Methane content major emission driver - Modified OPGEE model used	Winton, Queensland, Australia
Malek et al. (2022)	Advanced Geothermal System (AGS)	TEA, LCA, Sensitivity analysis	- CO₂ is more efficient than water - Deeper wells and multiple laterals reduce cost - Zero CO₂ emissions	Karlsruhe, Germany
Daniilidis et al. (2022)	High-Temperature Aquifer Thermal Energy Storage (HT-ATES)	TEA, LCA, Subsurface simulation	- System competitiveness improves with high transmissivity - CO₂ reduction up to 0.8 billion kg over 15 years - Low cut-off temp improves efficiency	Geneva, Switzerland
Aviles et al. (2021)	Hybrid Solar-Geothermal with Desalination	TEA, Modeling, Optimization	- Hybrid CST/PV system with geothermal heat improves efficiency - Lowest LCOE with 27.5% PV - Desalination feasible with minimal LCOE impact - Competitive water production cost	N/A (literature-based)
Bareiß et al. (2021)	PEM Water Electrolysis for H ₂ production	LCA, Scenario-based analysis	- PEMWE GWP reduced to 3.3 kg CO₂/kg H₂ with renewables - GHG reduction up to 75% - PEMWE supports grid and renewable integration	Germany

Bett & Jalilinasrabady (2021)	Geothermal Power	LCA	- Geothermal plants have low GHG (1.115-3.189 g CO₂/kWh) - Development phase is main source of emissions.	Unspecified
Stemmler et al. (2021)	Aquifer Thermal Energy Storage (ATES)	LCA, Monte Carlo Simulation	- ATES offers up to 97% GHG savings by 2050 - Key variables are cooling time and COP of heat pump.	Unspecified (modeled after U.S. data)
Fragiacomo & Genovese (2020)	PEM hydrogen production with wind, solar, and geothermal integration	TEA, LCA, Sensitivity Analysis	- Emissions analysis shows social and health benefits - Electricity price and demand scenarios significantly affect cost.	China
Sadeghi et al. (2020)	PV and Solar Thermal Electrolysis	TEA, LCA, Uncertainty Analysis	- GHG abatement cost: \$0.786/kg CO₂ (PV) and \$1.37/kg CO₂ (solar thermal) - PV more cost-effective for petroleum sector decarbonization.	Kenya
Wang et al. (2020)	Geothermal Power Systems (4 types)	LCA	Well drilling phase contributes most to environmental impact; higher geothermal gradients yield lower emissions.	Unspecified
Schuppler et al. (2019)	Aquifer Thermal Energy Storage (ATES)	TEA, LCA, Monte Carlo simulation, 30-year cost-benefit analysis	- ATES reduces electricity by 80% - Short 3-year payback - Saves 600 tCO₂/year - Provides detailed cost breakdowns	New Zealand
Valente et al. (2018)	Hydrogen production (various types)	LCA, Harmonization	- Introduced harmonised CEDnr protocol - CEDnr for renewable H₂: 3-184 MJ/kg vs. 201 MJ/kg for SMR - CEDnr strongly correlates with GWP	USA (California)

Despite this methodological breadth, the review highlights gaps in coverage across the value chain. The majority of hydrogen-focused studies still concentrate on the production phase, particularly via electrolysis, with limited exploration of subsurface hydrogen storage, especially in porous media such as aquifers. Only a few studies, such as [Yousefi et al. \(2023\)](#), address underground hydrogen storage in detail with combined TEA and LCA frameworks. A similar trend exists in the geothermal domain, where most research is centered on power production, while aquifer thermal energy storage (ATES) studies remain largely focused on heating and cooling applications, not electricity generation. Moreover, the ATES literature rarely integrates both TEA and LCA in a unified framework to evaluate the full system performance. Exceptions like [Daniilidis et al. \(2022\)](#) and [Schuppler et al. \(2019\)](#) begin to bridge this gap, but such examples are relatively scarce. Another notable development is the introduction of standardized LCA methodologies, as demonstrated by [Valente et al. \(2018\)](#), which can help facilitate cross-study comparisons and reduce methodological inconsistencies. Geographically, the studies span a broad range—from Europe and Australia to Asia and Africa—demonstrating global interest in decarbonization. However, site-specific factors like geothermal gradients, renewable electricity mix, and resource availability often dictate system feasibility and performance, reinforcing the need for localized analysis.

Overall, the literature demonstrates a growing trend toward combining TEA and LCA to evaluate clean energy technologies. However, comprehensive, integrated assessments of long-duration energy storage systems—particularly subsurface hydrogen and geothermal storage in porous media—remain limited. This highlights the need for future research that holistically examines both the economic viability and environmental impact of these emerging storage technologies under realistic, site-specific conditions.

Conclusions

Given the increasing renewable energy curtailment and the need for reliable seasonal storage solutions, this analysis combined real-world curtailment data with probabilistic Monte Carlo simulations and OAT sensitivity assessments to evaluate the technical performance, economic viability, and environmental impact

of two emerging long-duration storage technologies: (1) subsurface hydrogen storage and (2) synthetic geothermal storage. The key findings are summarized below:

Curtailment Trends

- Seasonal wind curtailment in ERCOT is highest during spring and fall, when wind speeds are elevated but electricity demand is relatively low. This seasonal imbalance highlights the need for long-duration storage to capture excess renewable generation and enhance grid utilization

Subsurface Hydrogen Storage

- System efficiency ranges between 40–53% under Monte Carlo simulation, demonstrating moderate robustness to input variability
- Curtailment availability is identified as the most influential factor for both system efficiency and leveled cost, as quantified by sensitivity analysis
- Electrolyzer efficiency has a strong effect on cost and emissions, with a $\pm 25\%$ change in efficiency resulting in $\pm 25\%$ change in LCOS and up to $\pm 8\%$ in emissions
- Reservoir-related factors, including withdrawal efficiency and injection volume, are also impactful, emphasizing the importance of subsurface design

Synthetic Geothermal Storage

- System efficiency is low but stable, between 12.0–15.3%, with an average of 13.34%
- Turbine and generator efficiency are the dominant parameters affecting system performance and emissions, underscoring the importance of conversion technologies
- Curtailment levels again play a critical role in economic and environmental outcomes, though the system shows lower sensitivity relative to hydrogen storage

Literature Review Insights

- Most studies prioritize hydrogen production and geothermal electricity generation, with limited focus on subsurface storage in porous formations
- Few assessments adopt integrated TEA–LCA frameworks for storage applications, especially in aquifer-based systems
- The adoption of harmonized LCA methodologies and advanced tools (e.g., optimization models) is critical for ensuring methodological consistency and enabling meaningful comparisons across studies

Future Outlook

The findings highlight that subsurface hydrogen storage offers greater potential for scalability, due to its higher system efficiency and stronger responsiveness to operational adjustments. In contrast, synthetic geothermal storage remains less efficient and would benefit significantly from technological innovations in heat-to-power conversion to enhance its overall performance and viability. Future research could focus on:

- Developing site-specific models that incorporate geological, climatic, and market factors
- Include policy and financial support (like carbon credits or storage incentives) in TEA–LCA evaluations
- Conducting comparative assessments across storage types to inform technology deployment in regions facing high renewable curtailment

By closing these gaps, the energy sector can accelerate the deployment of resilient, low-emission, and cost-effective long-duration storage systems—paving the way for greater integration of renewable energy into the grid.

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